

Subject Description Form

Subject Code	COMP5513
Subject Title	Financial Computing
Credit Value	3
Level	5
Pre-requisite/Exclusion	Nil
Objectives	The purpose of this subject is to provide an introduction to finance and to study the basic computational techniques/tools for finance. This includes the study of computational models and quantitative methods. After completing the subject, students will acquire: 1. fundamental concepts of finance in preparation for related studies 2. basic understanding of various computational techniques (e.g. mathematical methods and/or machine learning-based methods) which have been applied to solve problems in finance (e.g., options, stock price prediction, etc.) 3. experience in financial analysis
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. develop a thorough understanding of the key concepts of finance; b. apply various computational techniques for finance; c. solve related finance problems and evaluate the results; d. evaluate ethical issues in finance; e. practice presentation and communication skills (e.g., through case study report/presentations); and f. develop teamwork skills.
Subject Synopsis/ Indicative Syllabus	• Introduction to Finance: Finance Basics (e.g., Finance Industry, Time Value of Money), Debt and Bond, Equity, Derivatives, Other Investments (e.g., Foreign Exchange, Commodities, Hedge Funds), Related Theories (e.g., CAP-M). • Computational Methods for Finance: Mathematical, Statistical, Behavioural, Machine Learning-based, Simulation-based Methods etc. • Other Topics: FinTech (e.g., Cryptocurrencies), Ethics in Finance, Case Studies etc.
Teaching/Learning Methodology	Formal lectures will be used to present concepts about and mathematical models in finance and to introduce various computational techniques to solve financial computing problems. Students are expected to have a good mathematical background (e.g., knowledge of probability, statistics and calculus). Tutorials and/or labs will be used to discuss more details of computational techniques learned and to solve problems with models/tools. Seminars and/or guest lectures will be used for presentation of assignments and discussions (e.g., journal articles and conference papers on computational finance) as well as experience sharing

	(e.g., real-life experience).							
	39 hours of class activities including lecture, tutorial, workshop and/or seminar where applicable.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific Assessment Methods/Tasks	% weighting	Intended subject learning outcomes to be assessed					
			a	b	c	d	e	f
	Continuous Assessment: Individual Assignment(s), Group Project and Other Assessment(s)	30	✓	✓	✓	✓	✓	✓
	Final Examination	70	✓	✓	✓	✓		
	Total	100						
	The continuous assessment consists of three main components:							
• Individual Assessment: Evaluated through one or more individual assignments. • Group Assessment: Assessed via a group project, with consideration given to each student’s individual contributions. • Other Assessment(s) (e.g., participation): Assessed through in-class quizzes, etc.								
The group project is designed to assess all intended learning outcomes of the subject. The individual assignment(s) evaluate students’ knowledge and understanding of the subject matter. Other assessment(s) (e.g., in-class quizzes) are used to assess participation and monitor attendance. Finally, students’ overall performance is assessed through a formal examination.								
Note that, in accordance with departmental policy, the examination accounts for 70% of the overall grade. Equally important, active participation in lectures and engagement in learning activities are essential components of the learning process. Therefore, satisfactory completion of continuous assessment is mandatory. If a student fails the continuous assessment, the maximum overall grade attainable is D+, regardless of the examination result.								
For the general departmental policy on the use of generative artificial intelligence (GenAI), please refer to the Programme Requirement Document. GenAI may be used for assignment(s) and project(s), provided that the 3R framework is followed and the appropriate reporting mechanism, as outlined below, is observed:								
https://merlot.org/merlot/viewMaterial.htm?id=773417470								
Specific details and requirements will be provided in the assignment and project instructions. Note that the use of GenAI is strictly prohibited during the final examination.								

Student study effort expected	Class Contact:	
	Class activities (lecture, tutorial, workshop)	39 hours
	Other student study effort:	
	Assignment(s), Quizzes, Project, Examination	66 hours
	Total student study effort	105 hours
Reading list and references	<i>Books</i> 1. Bodie, Z., Kane, A., and Marcus, A., 2013, Investments, McGraw-Hill, 10th Edition. 2. Kosowski, R.L. and Neftci, S. N., 2015, Principles of Financial Engineering, Academic Press. 3. Hull, J. (2018) Options, futures, and other derivatives. Pearson Education, Inc. 4. Levy, G., 2016, Computational Finance using C and C#: derivatives and valuation, Elsevier. 5. Seydel, R.U. 2012, Tools for Computational Finance, Springer London. 6. Levy, H., Levy, M. and Solomon, S., 2003, Microscopic Simulation of Financial Markets, Academic Press. <i>Others</i> IEEE International Conference on Data Mining Asian Journal of Business and Information System Journal of Computational Intelligence in Finance Journal of Computational Finance	